–Supplemental Material–
Prune but Preserve: Decoupling Redundancy Suppression and Semantic Preservation for Efficient Time Series Clustering

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I. Baseline Descriptions

The baseline set is organized around the empirical question studied in the main paper, namely at which stage redundancy should be reduced for unsupervised MTS clustering. It covers five streams: full-sequence representation learning, input-side token exclusion, partial-observation clustering, structural subsequence indexing, and foundation-style time series modeling. This design separates the benefit of source-level patch admission from gains brought by contrastive objectives, spectral modeling, graph structure, incomplete-view learning, or general temporal pretraining. For embedding-output baselines, k-means gives final assignments with the label-provided cluster number k, following the same unsupervised protocol as the main paper.

FCACC [1] is a fuzzy cluster-aware contrastive clustering method for time series. It constructs augmented views and introduces cluster-aware hard negatives to reduce the gap between instance contrast and cluster assignment. FCACC represents a clustering-oriented full-sequence baseline, testing whether cluster-aware contrast can reduce background influence without source-level patch admission. The official implementation is available at FCACC.

TFMCC [2] performs time-frequency augmented multi-level contrastive clustering. Temporal and spectral augmentations enhance cluster-relevant patterns at different representation levels, but the complete sequence still enters the encoding pipeline. TFMCC tests whether temporal-spectral contrast can handle redundancy as effectively as pruning before encoder-side interaction. The official implementation is available at TFMCC.

GTM [3] learns general time series representations through frequency-domain attention and shared temporal modeling. As GTM is designed for transferable representation rather than clustering-specific reduction, k-means is applied to its embeddings for final assignment. The comparison examines whether broad temporal pretraining can replace explicit patch admission under redundancy-dominated MTS records. The official implementation is available at GTM.

EMTC [4] is the closest input-side exclusion baseline for MTS clustering. It suppresses redundancy through importance-aware variate-wise masking and enriches retained content with multi-endogenous views. Its comparison with SIGMA directly tests whether confidence-calibrated admission, routed semantic restoration, and prototype-guided refinement improve clustering after sparse token admission. The official implementation is available at EMTC.

FEI [5] introduces a frequency-domain information bottleneck by masking frequency components and predicting the corresponding embeddings. This design encourages informative spectral structure without manually defined positive and negative pairs. FEI serves as a spectral-reduction baseline, assessing whether patch-level admission still helps beyond frequency-oriented representation learning. The official implementation is available at FEI.

MVCIMTS [6] is a contrastive multi-view clustering method for MTS. It treats variates as views and learns complementary information through multi-view contrast. Since its partial observation comes from variables instead of temporal patches, MVCIMTS tests whether variable-wise incompleteness brings the same clustering benefit as temporal admission followed by routed restoration. The official implementation is available at MVCIMTS.

k-Graph [7] converts subsequences into graph representations and clusters graph-derived features. By building graphs across multiple subsequence lengths, k-Graph captures recurring local patterns and structural similarity. It provides a data-centric baseline for datasets whose clusters depend on shape correspondence or repeated subsequence structure. The official implementation is available at k-Graph.

TimesURL [8] is a universal self-supervised representation method for time series. It combines contrastive learning with temporal constraints to obtain general embeddings. In the prune-and-preserve comparison, TimesURL represents a

full-sequence self-supervised baseline because redundancy is not controlled through an admissible patch set. The official implementation is available at TimesURL.

UNITS [9] is a unified multi-task time series model. It represents task information as tokens in a shared modeling framework and learns transferable representations across domains, temporal scales, and sampling patterns. UNITS provides a foundation-style baseline for testing whether task-unified pretraining is sufficient for redundancy-dominated MTS clustering. Its comparison with SIGMA highlights the role of source-level admission and routed semantic preservation in cluster-oriented representation learning. The official implementation is available at UNITS.

II. Implementation Details

All experiments follow the unsupervised clustering protocol in the main paper. For each UEA dataset, training and test splits are concatenated before clustering, and labels are accessed only for metric computation. Each experiment is repeated with five random seeds, and the mean and standard deviation are reported. This protocol keeps the supplemental material consistent with the main-paper evaluation and makes the supplementary tables comparable to the representative results in Section IV of the full paper.

SIGMA uses Confidence-Calibrated Masked Admission as the default admission strategy. In the masking ablation, source-level admission is removed while all other settings remain unchanged, isolating the effect of controlling patch participation before encoder-side feature interaction. Unless otherwise specified, no input perturbation is added, and Gaussian noise is introduced only in the robustness study.

For patch-level admission, the patch length is set to $\ell_{\mathrm{p}} = 4$. Given $m = \lceil T/\ell_{\mathrm{p}} \rceil$ patches, the retained ratio is $r_{\mathrm{a}} = 0.5$, and the retained patch number is $m_{\mathrm{a}} = \lceil r_{\mathrm{a}} m \rceil$. Thus, the encoder processes a compact patch subset after admission. This setting reduces redundant patch participation before feature interaction while keeping enough temporal context for semantic restoration.

For endogenous-view representation learning, the number of views is $n_{\mathrm{v}} = 2$. Each view encoder uses hidden dimension $d_{\mathrm{h}} = 32$, and the fused clustering embedding has dimension $d_{\mathrm{z}} = 64$. The prototype alignment temperature is $\tau_{\mathrm{p}} = 0.1$. These values give SIGMA a compact representation budget while allowing admitted content to be organized into complementary latent views.

SIGMA is trained for $e_{\max} = 200$ epochs with mini-batch size $b = 64$. Adam is adopted with learning rate $\alpha = 0.001$. The loss weights are $\lambda_{\mathrm{rec}} = 1.0$, $\lambda_{\mathrm{orth}} = 0.1$, and $\lambda_{\mathrm{pa}} = 1.0$, corresponding to reconstruction, inter-view orthogonality, and prototype alignment, respectively. Reconstruction anchors temporal content after admission, orthogonality discourages view homogenization, and prototype alignment sharpens the fused cluster space. Under this configuration, admission control, semantic restoration, and cluster refinement keep distinct roles within the prune-and-preserve pipeline.

III. Supplementary Results

This section provides the full experimental record behind the condensed evaluation in the main paper. All results follow the unsupervised clustering protocol in Section II, so the added tables and figures serve as expanded checks rather than separate evaluation settings. The supplementary material includes complete 30-dataset results, Bonferroni-Dunn rank tests, detailed architecture and route ablations, loss-evolution curves, admission-based scalability tests, and additional robustness and compactness diagnostics.

The presentation follows the experimental questions in the main paper. Complete benchmark tables and rank tests extend Q1 from representative datasets to the full UEA benchmark. Architecture, route-internal, and newly added fine-grained mechanism ablations expand Q2 by tracing the contributions of admission, confidence calibration, endogenous-view generation, reconstruction routing, and prototype refinement at progressively finer levels. The additional CKA-based semantic preservation evaluation complements the prune-but-preserve analysis by examining whether compact representations retain representation-level structural information across representative clustering methods. Larger-scale and mask-rate tests further examine Q3, while the robustness diagnostics complement Q4 under noise perturbation and embedding compression. Together, these supplementary results provide more detailed evidence for the effectiveness, mechanism attribution, semantic preservation, efficiency, and robustness claims in the main paper without changing the evaluation protocol.

A. Complete Clustering Results on 30 UEA Datasets

TABLE I and TABLE II report the complete Q1 results on all 30 UEA datasets. ACC, F1, NMI, and ARI are computed under the same unsupervised protocol as the main paper. This subsection checks whether the trends reported on representative datasets extend to the full benchmark.

The complete records show that SIGMA performs well on datasets where cluster-related changes appear in compact fragments. For example, SIGMA reaches 0.9875 ACC and 0.9666 ARI on BasicMotions, obtains 0.8525 ACC and 0.7578 ARI on JapaneseVowels, and improves ACC and ARI to 0.6841 and 0.5305 on PenDigits. On SelfRegulationSCP1, SIGMA also gives the best ACC, F1, and ARI. These cases are consistent with the main-paper claim that reducing patch participation before encoding can preserve grouping cues.

The full tables also show that different datasets still favor different modeling biases. FCACC performs well on CharacterTrajectories, Cricket, NATOPS, and UWaveGestureLibrary. EMTC remains competitive on AtrialFibrillation, FingerMovements, HandMovementDirection, and MotorImagery. TFMCC leads several shape-sensitive or spectral-sensitive cases, including DuckDuckGeese, ERing, and RacketSports, and FEI performs well on Epilepsy, Libras, and PhonemeSpectra. Therefore, the full benchmark does not suggest that one design dominates every temporal pattern. Instead, Fig. 1 summarizes the aggregate behavior, where SIGMA obtains the lowest average rank on ACC, F1, NMI, and ARI. The complete tables

TABLE I

COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART I. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND. OOM DENOTES OUT-OF-MEMORY RUNS.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
ArticularyWordRecognition	ACC	0.7329 ± 0.0037	0.3555 ± 0.0380	0.3547 ± 0.0185	0.5030 ± 0.0253	0.4710 ± 0.0731	0.0497 ± 0.0124	0.2838 ± 0.0159	0.6929 ± 0.0516	0.2948 ± 0.0200	0.7603 ± 0.0232
	F1	0.8039 ± 0.0055	0.3526 ± 0.0348	0.3560 ± 0.0214	0.3994 ± 0.0241	0.4974 ± 0.0815	0.0109 ± 0.0096	0.2774 ± 0.0169	0.7425 ± 0.0413	0.3114 ± 0.0243	0.7929 ± 0.0392
	NMI	0.8408 ± 0.0064	0.4708 ± 0.0311	0.5070 ± 0.0272	0.7012 ± 0.0159	0.6185 ± 0.0744	0.0314 ± 0.0428	0.3948 ± 0.0099	0.8215 ± 0.0145	0.4280 ± 0.0198	0.8276 ± 0.0165
	ARI	0.6641 ± 0.0045	0.1833 ± 0.0296	0.1615 ± 0.0271	0.4466 ± 0.0210	0.3352 ± 0.0843	0.0019 ± 0.0031	0.1224 ± 0.0091	0.6214 ± 0.0340	0.1395 ± 0.0188	0.6466 ± 0.0257
AtrialFibrillation	ACC	0.4333 ± 0.0000	0.4667 ± 0.0422	0.6133 ± 0.0499	0.4333 ± 0.0558	0.3867 ± 0.0163	0.4095 ± 0.0440	0.4400 ± 0.0389	0.4600 ± 0.0490	0.4500 ± 0.0167	0.5000 ± 0.0365
	F1	0.4303 ± 0.0028	0.4593 ± 0.0393	0.5828 ± 0.0633	0.4041 ± 0.0191	0.3543 ± 0.0194	0.3491 ± 0.0555	0.4295 ± 0.0406	0.4169 ± 0.0278	0.4343 ± 0.0089	0.4956 ± 0.0778
	NMI	0.1005 ± 0.0040	0.0726 ± 0.0277	0.2883 ± 0.0758	0.0544 ± 0.0351	0.0353 ± 0.0237	0.1445 ± 0.0821	0.1143 ± 0.0748	0.0858 ± 0.0339	0.0643 ± 0.0072	0.1251 ± 0.0390
	ARI	-0.0056 ± 0.0011	0.0051 ± 0.0258	0.1248 ± 0.0788	0.0064 ± 0.0282	-0.0360 ± 0.0089	0.1058 ± 0.0600	0.0039 ± 0.0462	0.0136 ± 0.0342	-0.0125 ± 0.0012	0.0582 ± 0.0393
BasicMotions	ACC	0.6675 ± 0.0100	0.7300 ± 0.1086	0.7950 ± 0.0292	0.9850 ± 0.0122	0.6775 ± 0.0443	0.2975 ± 0.0827	0.8875 ± 0.1321	0.7700 ± 0.1582	0.5437 ± 0.0812	0.9875 ± 0.0079
	F1	0.8228 ± 0.0098	0.7311 ± 0.1060	0.7832 ± 0.0275	0.9701 ± 0.0235	0.7351 ± 0.0723	0.1517 ± 0.0794	0.8792 ± 0.1500	0.8627 ± 0.0835	0.5705 ± 0.0412	0.9875 ± 0.0079
	NMI	0.8048 ± 0.0020	0.5569 ± 0.1633	0.6949 ± 0.0711	0.9627 ± 0.0248	0.6792 ± 0.0317	0.1007 ± 0.1775	0.8626 ± 0.0575	0.8487 ± 0.0634	0.3898 ± 0.1107	0.9665 ± 0.0191
	ARI	0.6295 ± 0.0045	0.4526 ± 0.1902	0.6085 ± 0.0606	0.9606 ± 0.0311	0.5317 ± 0.0351	0.0690 ± 0.1381	0.8174 ± 0.1218	0.7349 ± 0.1448	0.2989 ± 0.1276	0.9666 ± 0.0208
CharacterTrajectories	ACC	0.7981 ± 0.0012	0.0651 ± 0.0000	0.3000 ± 0.0261	0.3677 ± 0.0448	0.6040 ± 0.0096	0.0866 ± 0.0318	0.4500 ± 0.0167	0.0651 ± 0.0000	0.4533 ± 0.0267	0.7668 ± 0.0529
	F1	0.8583 ± 0.0012	0.0061 ± 0.0000	0.2886 ± 0.0434	0.3405 ± 0.0281	0.6289 ± 0.0099	0.0196 ± 0.0166	0.4350 ± 0.0149	0.1222 ± 0.0000	0.1703 ± 0.0373	0.8019 ± 0.0378
	NMI	0.8726 ± 0.0013	0.0000 ± 0.0000	0.3633 ± 0.0432	0.5756 ± 0.0418	0.7095 ± 0.0041	0.0682 ± 0.1107	0.5347 ± 0.0121	0.0000 ± 0.0000	0.1776 ± 0.0688	0.8261 ± 0.0315
	ARI	0.7705 ± 0.0012	0.0000 ± 0.0000	0.1756 ± 0.0296	0.1913 ± 0.0474	0.5137 ± 0.0138	0.0181 ± 0.0345	0.3211 ± 0.0193	0.0000 ± 0.0000	0.0532 ± 0.0279	0.6933 ± 0.0563
Cricket	ACC	0.9407 ± 0.0288	0.4533 ± 0.0264	0.5528 ± 0.0204	0.6922 ± 0.0664	0.5744 ± 0.0834	0.1167 ± 0.0350	0.4589 ± 0.0218	0.6722 ± 0.0813	0.4922 ± 0.0990	0.8489 ± 0.0787
	F1	0.9599 ± 0.0024	0.4393 ± 0.0376	0.5635 ± 0.0383	0.6319 ± 0.0703	0.6027 ± 0.0590	0.0417 ± 0.0250	0.4545 ± 0.0225	0.7562 ± 0.0557	0.5414 ± 0.0992	0.8838 ± 0.0523
	NMI	0.9509 ± 0.0020	0.5189 ± 0.0101	0.6181 ± 0.0286	0.7858 ± 0.0464	0.6794 ± 0.0579	0.0817 ± 0.0964	0.4752 ± 0.0228	0.7958 ± 0.0391	0.5947 ± 0.0841	0.8764 ± 0.0455
	ARI	0.9051 ± 0.0158	0.2584 ± 0.0194	0.2971 ± 0.0365	0.5842 ± 0.0827	0.4468 ± 0.0872	0.0171 ± 0.0299	0.2533 ± 0.0250	0.6028 ± 0.0783	0.3359 ± 0.1072	0.7672 ± 0.0967
DuckDuckGeese	ACC	0.3067 ± 0.0125	0.4140 ± 0.0162	0.4240 ± 0.0408	0.4720 ± 0.0172	0.3220 ± 0.0117	0.2620 ± 0.0519	0.3280 ± 0.0160	0.3220 ± 0.0147	0.2960 ± 0.0136	0.4020 ± 0.0325
	F1	0.2611 ± 0.0200	0.4042 ± 0.0246	0.4003 ± 0.0494	0.3394 ± 0.0193	0.3013 ± 0.0347	0.1650 ± 0.0575	0.3017 ± 0.0107	0.2641 ± 0.0182	0.2869 ± 0.0075	0.3819 ± 0.0291
	NMI	0.0800 ± 0.0179	0.2483 ± 0.0205	0.2082 ± 0.0380	0.2791 ± 0.0267	0.1030 ± 0.0090	0.0335 ± 0.0301	0.0958 ± 0.0101	0.1732 ± 0.0351	0.0682 ± 0.0175	0.1718 ± 0.0577
	ARI	0.0227 ± 0.0109	0.1394 ± 0.0210	0.0729 ± 0.0331	0.1780 ± 0.0248	0.0369 ± 0.0091	0.0073 ± 0.0170	0.0225 ± 0.0087	0.0375 ± 0.0110	0.0064 ± 0.0105	0.1000 ± 0.0580
EigenWorms	ACC	OOM	0.3876 ± 0.0072	OOM	OOM	0.4185 ± 0.0361	0.4324 ± 0.0024	0.3135 ± 0.0236	0.3660 ± 0.0510	0.4525 ± 0.0366	0.4332 ± 0.0402
	F1	OOM	0.2868 ± 0.0068	OOM	OOM	0.3899 ± 0.0605	0.1545 ± 0.0173	0.2920 ± 0.0180	0.3035 ± 0.0806	0.4410 ± 0.0601	0.4016 ± 0.0483
	NMI	OOM	0.0683 ± 0.0035	OOM	OOM	0.1826 ± 0.0152	0.0300 ± 0.0094	0.0848 ± 0.0181	0.0697 ± 0.0572	0.2872 ± 0.0919	0.1885 ± 0.0300
	ARI	OOM	0.0626 ± 0.0052	OOM	OOM	0.1453 ± 0.0227	0.0098 ± 0.0046	0.0400 ± 0.0160	0.0515 ± 0.0699	0.1694 ± 0.0486	0.1620 ± 0.0458
Epilepsy	ACC	0.4279 ± 0.0297	0.3869 ± 0.0404	0.5203 ± 0.0244	0.4953 ± 0.0477	0.6364 ± 0.0808	0.2953 ± 0.0324	0.6167 ± 0.0316	0.5156 ± 0.0189	0.4982 ± 0.0646	0.4756 ± 0.0321
	F1	0.4295 ± 0.0302	0.3710 ± 0.0430	0.5136 ± 0.0265	0.4147 ± 0.0552	0.6200 ± 0.0821	0.1882 ± 0.0235	0.6200 ± 0.0260	0.4979 ± 0.0080	0.5086 ± 0.0930	0.4724 ± 0.0357
	NMI	0.1397 ± 0.0347	0.1015 ± 0.0345	0.2476 ± 0.0419	0.3029 ± 0.0794	0.4079 ± 0.0773	0.0205 ± 0.0340	0.4013 ± 0.0415	0.3363 ± 0.0795	0.3126 ± 0.0851	0.2652 ± 0.0289
	ARI	0.0960 ± 0.0211	0.0814 ± 0.0434	0.1803 ± 0.0389	0.1992 ± 0.0638	0.3829 ± 0.0804	0.0076 ± 0.0184	0.3275 ± 0.0526	0.2288 ± 0.0542	0.2320 ± 0.0733	0.1889 ± 0.0197
ERing	ACC	0.6327 ± 0.0360	0.4180 ± 0.0585	0.6704 ± 0.0486	0.8607 ± 0.0218	0.7120 ± 0.0918	0.3667 ± 0.1560	0.5380 ± 0.0421	0.7433 ± 0.1161	0.4087 ± 0.0639	0.7980 ± 0.0749
	F1	0.6306 ± 0.0393	0.4117 ± 0.0586	0.6741 ± 0.0496	0.7564 ± 0.0332	0.7508 ± 0.0811	0.2682 ± 0.1646	0.5469 ± 0.0403	0.7862 ± 0.0839	0.4159 ± 0.0698	0.8003 ± 0.0701
	NMI	0.5660 ± 0.0104	0.2357 ± 0.0481	0.5487 ± 0.0473	0.7821 ± 0.0321	0.6562 ± 0.0772	0.3054 ± 0.2115	0.3843 ± 0.0143	0.7407 ± 0.0313	0.2528 ± 0.0802	0.6872 ± 0.0892
	ARI	0.4231 ± 0.0204	0.1602 ± 0.0466	0.4382 ± 0.0499	0.7051 ± 0.0415	0.5669 ± 0.0959	0.1897 ± 0.1599	0.2816 ± 0.0319	0.6467 ± 0.0757	0.1578 ± 0.0647	0.6168 ± 0.1270
EthanolConcentration	ACC	0.2992 ± 0.0025	0.2786 ± 0.0051	0.3605 ± 0.0224	0.2634 ± 0.0096	0.2901 ± 0.0017	0.2962 ± 0.0172	0.2920 ± 0.0100	0.2962 ± 0.0044	0.2691 ± 0.0034	0.2969 ± 0.0153
	F1	0.2759 ± 0.0027	0.2529 ± 0.0093	0.3458 ± 0.0279	0.4681 ± 0.0189	0.2744 ± 0.0087	0.2442 ± 0.0306	0.2882 ± 0.0110	0.2747 ± 0.0067	0.2090 ± 0.0059	0.2732 ± 0.0102
	NMI	0.0149 ± 0.0009	0.0047 ± 0.0010	0.0574 ± 0.0189	0.0124 ± 0.0098	0.0116 ± 0.0018	0.0176 ± 0.0112	0.0074 ± 0.0029	0.0199 ± 0.0025	0.0025 ± 0.0015	0.0190 ± 0.0157
	ARI	0.0012 ± 0.0003	-0.0013 ± 0.0010	0.0307 ± 0.0121	0.0003 ± 0.0007	0.0032 ± 0.0005	0.0072 ± 0.0073	0.0008 ± 0.0024	0.0039 ± 0.0009	-0.0014 ± 0.0010	0.0097 ± 0.0119
FaceDetection	ACC	0.5130 ± 0.0030	OOM	0.5146 ± 0.0047	0.5051 ± 0.0029	0.5077 ± 0.0031	OOM	0.5038 ± 0.0018	0.5001 ± 0.0000	0.5054 ± 0.0040	0.5198 ± 0.0044
	F1	0.5125 ± 0.0028	OOM	0.4808 ± 0.0369	0.6113 ± 0.0585	0.5051 ± 0.0039	OOM	0.5028 ± 0.0023	0.3336 ± 0.0000	0.5030 ± 0.0060	0.5192 ± 0.0046
	NMI	0.0005 ± 0.0002	OOM	0.0010 ± 0.0004	0.0002 ± 0.0002	0.0002 ± 0.0001	OOM	0.0001 ± 0.0000	0.0027 ± 0.0000	0.0001 ± 0.0002	0.0012 ± 0.0005
	ARI	0.0006 ± 0.0003	OOM	0.0007 ± 0.0006	0.0001 ± 0.0001	0.0002 ± 0.0002	OOM	-0.0000 ± 0.0000	0.0000 ± 0.0000	0.0001 ± 0.0003	0.0015 ± 0.0006
FingerMovements	ACC	0.5250 ± 0.0052	0.5183 ± 0.0085	0.5680 ± 0.0160	0.5072 ± 0.0055	0.5163 ± 0.0127	0.5260 ± 0.0041	0.5264 ± 0.0026	0.5192 ± 0.0082	0.5130 ± 0.0054	0.5221 ± 0.0129
	F1	0.4951 ± 0.0111	0.4982 ± 0.0212	0.5349 ± 0.0218	0.6004 ± 0.0599	0.5051 ± 0.0184	0.5230 ± 0.0023	0.4997 ± 0.0023	0.5161 ± 0.0058	0.5124 ± 0.0051	0.5183 ± 0.0074
	NMI	0.0027 ± 0.0009	0.0014 ± 0.0009	0.0210 ± 0.0132	0.0023 ± 0.0035	0.0013 ± 0.0016	0.0021 ± 0.0007	0.0028 ± 0.0005	0.0013 ± 0.0009	0.0006 ± 0.0003	0.0022 ± 0.0026
	ARI	0.0008 ± 0.0009	-0.0004 ± 0.0009	0.0121 ± 0.0087	-0.0010 ± 0.0008	-0.0005 ± 0.0021	0.0004 ± 0.0009	0.0009 ± 0.0005	-0.0006 ± 0.0013	-0.0016 ± 0.0004	0.0003 ± 0.0031
HandMovementDirection	ACC	0.2829 ± 0.0063	0.2957 ± 0.0087	0.4405 ± 0.0359	0.3248 ± 0.0164	0.3017 ± 0.0088	0.3017 ± 0.0123	0.3026 ± 0.0099	0.3120 ± 0.0097	0.3743 ± 0.0375	0.3179 ± 0.0100
	F1	0.2108 ± 0.0397	0.2907 ± 0.0061	0.4070 ± 0.0226	0.3328 ± 0.0189	0.2867 ± 0.0112	0.2667 ± 0.0205	0.2996 ± 0.0128	0.3089 ± 0.0099	0.3667 ± 0.0383	0.3145 ± 0.0104
	NMI	0.0168 ± 0.0104	0.0156 ± 0.0061	0.0910 ± 0.0245	0.0286 ± 0.0110	0.0184 ± 0.0049	0.0206 ± 0.0103	0.0144 ± 0.0047	0.0208 ± 0.0045	0.0818 ± 0.0394	0.0208 ± 0.0050
	ARI	-0.0047 ± 0.0038	0.0015 ± 0.0062	0.0475 ± 0.0166	0.0109 ± 0.0104	0.0001 ± 0.0045	0.0007 ± 0.0032	0.0003 ± 0.0032	0.0046 ± 0.0035	0.0488 ± 0.0282	0.0052 ± 0.0047
Handwriting	ACC	0.2433 ± 0.0041	0.1264 ± 0.0027	0.1536 ± 0.0097	0.1090 ± 0.0209	0.1778 ± 0.0097	0.0878 ± 0.0160	0.1708 ± 0.0101	0.1488 ± 0.0067	0.2940 ± 0.0068	0.2530 ± 0.0126
	F1	0.2418 ± 0.0080	0.1260 ± 0.0020	0.1501 ± 0.0114	0.1206 ± 0.0171	0.1741 ± 0.0071	0.0352 ± 0.0127	0.1679 ± 0.0098	0.1484 ± 0.0078	0.2881 ± 0.0075	0.2591 ± 0.0132
	NMI	0.3723 ± 0.0061	0.1523 ± 0.0052	0.1934 ± 0.0070	0.1228 ± 0.0510	0.2497 ± 0.0235	0.0637 ± 0.0274	0.2260 ± 0.0097	0.2009 ± 0.0035	0.0101 ± 0.0049	0.3652 ± 0.0139
	ARI	0.1106 ± 0.0035	0.0125 ± 0.0016	0.0254 ± 0.0036	0.0195 ± 0.0090	0.0494 ± 0.0107	0.0084 ± 0.0048	0.0429 ± 0.0045	0.0282 ± 0.0012	-0.0047 ± 0.0030	0.1134 ± 0.0113

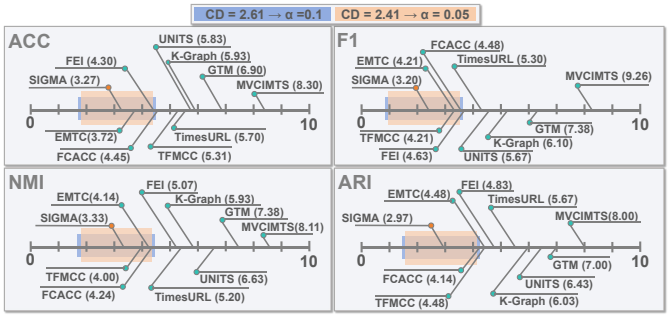

Fig. 1. Bonferroni-Dunn rank test on 30 UEA datasets under ACC, F1, NMI, and ARI. Lower average rank indicates stronger clustering quality, and critical-difference bars report significance thresholds at $\alpha = 0.1$ and $\alpha = 0.05$.

and the rank test together show that the main-paper results are not limited to selected examples.

B. Complete Architecture and Route Ablations

The full paper reports averaged ablation results for Q2, while TABLEs III–VI provide dataset-level results on all 30 UEA datasets under the protocol in Section II. The architecture ablations remove CCMA and MEVG separately or jointly, whereas the route ablations remove reconstruction and contrastive refinement in the same manner. These controlled variants examine whether the components provide distinct benefits across heterogeneous temporal patterns and whether their combination is necessary for stable clustering.

TABLEs III and IV show that the contribution of CCMA and MEVG varies with dataset characteristics. A reduced variant can occasionally perform better on an individual metric, indicating that different motion, speech, sensor, and biomedical datasets depend differently on admission control and endogenous-view representation. The aggregate trend, however, favors their coordination. For example, on UWaveGestureLibrary, the complete model obtains an ARI of 0.1996, compared with 0.0954 without CCMA, 0.1487 without MEVG, and 0.0860 when both are removed. More broadly,

TABLE II

COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART II. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	*k*-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
Heartbeat	ACC	0.5844 ± 0.0079	0.5340 ± 0.0292	0.7180 ± 0.0020	0.7076 ± 0.0476	0.7090 ± 0.0000	0.6758 ± 0.0497	0.7296 ± 0.0025	0.7149 ± 0.0106	0.6244 ± 0.0238	0.6817 ± 0.0546
	F1	0.5688 ± 0.0055	0.4974 ± 0.0193	0.7553 ± 0.1598	0.6934 ± 0.0669	0.8298 ± 0.0000	0.4592 ± 0.0483	0.5461 ± 0.0071	0.7922 ± 0.0760	0.5347 ± 0.0192	0.5944 ± 0.0367
	NMI	0.0389 ± 0.0016	0.0012 ± 0.0015	0.0106 ± 0.0023	0.0465 ± 0.0073	0.0204 ± 0.0000	0.0016 ± 0.0020	0.0344 ± 0.0041	0.0300 ± 0.0207	0.0070 ± 0.0077	0.0382 ± 0.0220
	ARI	0.0258 ± 0.0062	0.0030 ± 0.0062	-0.0029 ± 0.0061	0.0966 ± 0.0306	-0.0144 ± 0.0000	0.0018 ± 0.0101	0.0875 ± 0.0071	0.0220 ± 0.0714	0.0243 ± 0.0111	0.0981 ± 0.0485
InsectWingbeat	ACC	0.1651 ± 0.0020	0.1003 ± 0.0000	0.1251 ± 0.0049	0.1099 ± 0.0024	0.1533 ± 0.0004	OOM	0.1451 ± 0.0010	0.1000 ± 0.0000	0.0872 ± 0.0071	0.1336 ± 0.0017
	F1	0.1623 ± 0.0020	0.0189 ± 0.0001	0.0945 ± 0.0222	0.0962 ± 0.0074	0.1062 ± 0.0051	OOM	0.1374 ± 0.0049	0.1818 ± 0.0000	0.0812 ± 0.0055	0.1209 ± 0.0043
	NMI	0.0428 ± 0.0011	0.0010 ± 0.0001	0.0090 ± 0.0015	0.1883 ± 0.0060	0.0339 ± 0.0008	OOM	0.0210 ± 0.0008	0.0000 ± 0.0000	0.0032 ± 0.0061	0.0146 ± 0.0015
	ARI	0.0216 ± 0.0005	0.0000 ± 0.0000	0.0027 ± 0.0013	0.0377 ± 0.0055	0.0130 ± 0.0007	OOM	0.0087 ± 0.0002	0.0000 ± 0.0000	0.0043 ± 0.0000	0.0059 ± 0.0006
JapaneseVowels	ACC	0.6286 ± 0.0530	0.1844 ± 0.0000	0.3203 ± 0.0273	0.4766 ± 0.0639	0.5198 ± 0.0223	0.1844 ± 0.0000	0.3369 ± 0.0199	0.1844 ± 0.0000	0.2619 ± 0.0271	0.8525 ± 0.0431
	F1	0.6617 ± 0.0303	0.0346 ± 0.0000	0.3325 ± 0.0365	0.4274 ± 0.0355	0.5551 ± 0.0053	0.0346 ± 0.0000	0.3069 ± 0.0236	0.3113 ± 0.0000	0.2755 ± 0.0282	0.8494 ± 0.0383
	NMI	0.5950 ± 0.0134	0.0000 ± 0.0000	0.2410 ± 0.0418	0.5250 ± 0.0397	0.4727 ± 0.0084	0.0000 ± 0.0000	0.2820 ± 0.0112	0.0000 ± 0.0000	0.1643 ± 0.0209	0.8113 ± 0.0180
	ARI	0.4771 ± 0.0389	0.0000 ± 0.0000	0.1269 ± 0.0412	0.3167 ± 0.0466	0.3049 ± 0.0281	0.0000 ± 0.0000	0.1511 ± 0.0210	0.0000 ± 0.0000	0.0648 ± 0.0171	0.7578 ± 0.0423
Libras	ACC	0.3822 ± 0.0052	0.3894 ± 0.0321	0.4011 ± 0.0315	0.3544 ± 0.0099	0.4317 ± 0.0256	0.1456 ± 0.0416	0.3844 ± 0.0208	0.2700 ± 0.0148	0.2839 ± 0.0116	0.3711 ± 0.0125
	F1	0.3931 ± 0.0193	0.3772 ± 0.0369	0.4028 ± 0.0406	0.2626 ± 0.0226	0.4438 ± 0.0298	0.0627 ± 0.0302	0.3700 ± 0.0242	0.2767 ± 0.0166	0.2853 ± 0.0157	0.3655 ± 0.0158
	NMI	0.5311 ± 0.0095	0.4697 ± 0.0327	0.4890 ± 0.0322	0.4698 ± 0.0245	0.5056 ± 0.0189	0.1311 ± 0.0783	0.5126 ± 0.0062	0.3101 ± 0.0185	0.3562 ± 0.0152	0.4806 ± 0.0259
	ARI	0.2484 ± 0.0082	0.2143 ± 0.0276	0.1881 ± 0.0295	0.2019 ± 0.0228	0.2423 ± 0.0142	0.0370 ± 0.0255	0.2532 ± 0.0127	0.0901 ± 0.0166	0.1246 ± 0.0146	0.2089 ± 0.0196
LSST	ACC	0.3434 ± 0.0022	0.1224 ± 0.0038	0.3672 ± 0.0074	0.2491 ± 0.0169	0.3327 ± 0.0037	0.3227 ± 0.0092	0.1578 ± 0.0113	0.3293 ± 0.0025	0.1805 ± 0.0098	0.2019 ± 0.0029
	F1	0.1918 ± 0.0134	0.0984 ± 0.0072	0.2055 ± 0.0297	0.2114 ± 0.0148	0.1689 ± 0.0141	0.0521 ± 0.0222	0.1149 ± 0.0053	0.2387 ± 0.0157	0.1376 ± 0.0084	0.1840 ± 0.0147
	NMI	0.1188 ± 0.0020	0.0311 ± 0.0056	0.1959 ± 0.0261	0.2066 ± 0.0198	0.1338 ± 0.0040	0.0237 ± 0.0303	0.0764 ± 0.0061	0.1398 ± 0.0038	0.0543 ± 0.0132	0.1396 ± 0.0043
	ARI	0.0453 ± 0.0007	0.0096 ± 0.0022	0.0936 ± 0.0182	0.0634 ± 0.0311	0.0859 ± 0.0106	0.0070 ± 0.0089	0.0222 ± 0.0040	0.0254 ± 0.0061	0.0207 ± 0.0068	0.0549 ± 0.0028
MotorImagery	ACC	0.5016 ± 0.0013	0.5360 ± 0.0147	0.5840 ± 0.0258	0.5048 ± 0.0072	0.5058 ± 0.0035	0.5111 ± 0.0059	0.5159 ± 0.0111	0.5127 ± 0.0048	0.5333 ± 0.0173	0.5508 ± 0.0070
	F1	0.3512 ± 0.0022	0.5084 ± 0.0319	0.5308 ± 0.0377	0.6586 ± 0.0706	0.3794 ± 0.0205	0.4565 ± 0.0416	0.5093 ± 0.0107	0.4401 ± 0.0338	0.5180 ± 0.0271	0.5362 ± 0.0175
	NMI	0.0003 ± 0.0002	0.0055 ± 0.0031	0.0540 ± 0.0291	0.0023 ± 0.0040	0.0011 ± 0.0008	0.0017 ± 0.0016	0.0012 ± 0.0011	0.0012 ± 0.0004	0.0044 ± 0.0042	0.0096 ± 0.0034
	ARI	-0.0002 ± 0.0000	0.0040 ± 0.0035	0.0254 ± 0.0160	-0.0003 ± 0.0004	-0.0003 ± 0.0001	-0.0010 ± 0.0011	-0.0010 ± 0.0014	-0.0006 ± 0.0001	0.0033 ± 0.0059	0.0082 ± 0.0028
NATOPS	ACC	0.6537 ± 0.0975	0.4861 ± 0.0512	0.3978 ± 0.0514	0.5672 ± 0.0859	0.4350 ± 0.0268	0.3267 ± 0.0919	0.3817 ± 0.0194	0.3522 ± 0.0811	0.2972 ± 0.0249	0.5022 ± 0.0157
	F1	0.6529 ± 0.0968	0.4643 ± 0.0539	0.4342 ± 0.1010	0.5167 ± 0.0816	0.4372 ± 0.0599	0.2052 ± 0.0868	0.3871 ± 0.0168	0.2131 ± 0.0691	0.2951 ± 0.0249	0.4958 ± 0.0103
	NMI	0.6454 ± 0.0805	0.4355 ± 0.0551	0.2990 ± 0.1190	0.5575 ± 0.0724	0.3896 ± 0.0459	0.2876 ± 0.1673	0.1897 ± 0.0207	0.4115 ± 0.1613	0.1097 ± 0.0324	0.5159 ± 0.0230
	ARI	0.5091 ± 0.1162	0.3056 ± 0.0590	0.1759 ± 0.0826	0.4179 ± 0.0983	0.2563 ± 0.0459	0.1603 ± 0.1191	0.1147 ± 0.0214	0.2551 ± 0.1261	0.0549 ± 0.0222	0.3489 ± 0.0284
PEMS-SF	ACC	0.4864 ± 0.0558	0.1968 ± 0.0065	0.4301 ± 0.0842	0.2268 ± 0.0179	0.3582 ± 0.0366	0.2514 ± 0.0897	0.4268 ± 0.0172	0.2377 ± 0.0102	0.4409 ± 0.0466	0.4436 ± 0.0844
	F1	0.5110 ± 0.0447	0.1941 ± 0.0064	0.4202 ± 0.0807	0.2263 ± 0.0204	0.3782 ± 0.0333	0.1185 ± 0.0734	0.4239 ± 0.0214	0.2370 ± 0.0154	0.4683 ± 0.0591	0.4902 ± 0.1198
	NMI	0.4510 ± 0.0416	0.0348 ± 0.0119	0.3687 ± 0.1015	0.1270 ± 0.0168	0.3336 ± 0.0281	0.2715 ± 0.2226	0.4593 ± 0.0196	0.1742 ± 0.0226	0.4167 ± 0.0441	0.4570 ± 0.1104
	ARI	0.2911 ± 0.0478	0.0030 ± 0.0034	0.2134 ± 0.0924	0.0374 ± 0.0156	0.1673 ± 0.0312	0.1333 ± 0.1105	0.2557 ± 0.0121	0.0651 ± 0.0082	0.2471 ± 0.0435	0.2839 ± 0.0947
PenDigits	ACC	0.4246 ± 0.0409	0.4128 ± 0.0672	0.4089 ± 0.0499	0.6548 ± 0.0159	0.3484 ± 0.0354	0.2670 ± 0.0505	0.4078 ± 0.0162	0.3647 ± 0.0150	0.4687 ± 0.0517	0.6841 ± 0.0284
	F1	0.4116 ± 0.0303	0.3971 ± 0.0746	0.4086 ± 0.0578	0.5693 ± 0.0171	0.3548 ± 0.0321	0.1710 ± 0.0688	0.3825 ± 0.0213	0.3747 ± 0.0150	0.4542 ± 0.0512	0.6938 ± 0.0411
	NMI	0.3233 ± 0.0199	0.3405 ± 0.0616	0.3994 ± 0.0617	0.6583 ± 0.0086	0.3059 ± 0.0165	0.2076 ± 0.0559	0.3390 ± 0.0131	0.2611 ± 0.0165	0.4348 ± 0.0630	0.6485 ± 0.0189
	ARI	0.2218 ± 0.0289	0.2151 ± 0.0560	0.2313 ± 0.0515	0.5196 ± 0.0193	0.1503 ± 0.0122	0.1033 ± 0.0425	0.2183 ± 0.0158	0.1528 ± 0.0112	0.2822 ± 0.0658	0.5305 ± 0.0276
PhonemeSpectra	ACC	0.1149 ± 0.0003	0.0633 ± 0.0006	0.0963 ± 0.0032	0.1124 ± 0.0038	0.1283 ± 0.0024	0.0454 ± 0.0059	0.0916 ± 0.0028	0.0871 ± 0.0023	0.0812 ± 0.0071	0.1127 ± 0.0015
	F1	0.1159 ± 0.0004	0.0605 ± 0.0010	0.0944 ± 0.0004	0.1048 ± 0.0067	0.1242 ± 0.0038	0.0099 ± 0.0028	0.0913 ± 0.0033	0.0845 ± 0.0017	0.0786 ± 0.0080	0.1126 ± 0.0017
	NMI	0.1793 ± 0.0023	0.0451 ± 0.0015	0.1386 ± 0.0089	0.2039 ± 0.0025	0.2085 ± 0.0038	0.0366 ± 0.0136	0.1090 ± 0.0036	0.1174 ± 0.0070	0.0932 ± 0.0191	0.1758 ± 0.0026
	ARI	0.0339 ± 0.0007	0.0038 ± 0.0003	0.0166 ± 0.0040	0.0424 ± 0.0048	0.0449 ± 0.0011	0.0046 ± 0.0026	0.0178 ± 0.0015	0.0186 ± 0.0020	0.0143 ± 0.0045	0.0334 ± 0.0008
RacketSports	ACC	0.4172 ± 0.0058	0.3921 ± 0.0150	0.4355 ± 0.0286	0.6205 ± 0.0400	0.4257 ± 0.0251	0.3129 ± 0.0296	0.3551 ± 0.0324	0.5036 ± 0.0475	0.3743 ± 0.0375	0.6007 ± 0.0721
	F1	0.4138 ± 0.0065	0.3898 ± 0.0135	0.4064 ± 0.0374	0.5330 ± 0.0327	0.4067 ± 0.0328	0.2105 ± 0.0508	0.3529 ± 0.0340	0.5273 ± 0.0345	0.3667 ± 0.0383	0.6006 ± 0.0707
	NMI	0.1398 ± 0.0042	0.1024 ± 0.0386	0.1138 ± 0.0264	0.4687 ± 0.0339	0.1250 ± 0.0289	0.0688 ± 0.0371	0.0601 ± 0.0348	0.3279 ± 0.0420	0.0818 ± 0.0394	0.3488 ± 0.0845
	ARI	0.0880 ± 0.0042	0.0590 ± 0.0167	0.0776 ± 0.0307	0.3723 ± 0.0445	0.0924 ± 0.0200	0.0127 ± 0.0191	0.0400 ± 0.0293	0.2394 ± 0.0441	0.0488 ± 0.0282	0.2858 ± 0.0965
SelfRegulationSCP1	ACC	0.5362 ± 0.0031	0.5447 ± 0.0307	0.7126 ± 0.0431	0.5027 ± 0.0000	0.6096 ± 0.0472	0.5537 ± 0.0342	0.5586 ± 0.0087	0.7340 ± 0.0059	0.5811 ± 0.0402	0.7754 ± 0.0128
	F1	0.5081 ± 0.0040	0.5434 ± 0.0324	0.7073 ± 0.0419	0.7065 ± 0.0000	0.6082 ± 0.0487	0.4723 ± 0.0964	0.5584 ± 0.0087	0.7184 ± 0.0076	0.5676 ± 0.0372	0.7745 ± 0.0125
	NMI	0.0059 ± 0.0009	0.0086 ± 0.0093	0.1642 ± 0.0748	0.0000 ± 0.0000	0.0421 ± 0.0290	0.0261 ± 0.0139	0.0102 ± 0.0031	0.2493 ± 0.0044	0.0328 ± 0.0305	0.2382 ± 0.0246
	ARI	0.0039 ± 0.0009	0.0100 ± 0.0128	0.1857 ± 0.0753	0.0000 ± 0.0000	0.0553 ± 0.0393	0.0154 ± 0.0124	0.0123 ± 0.0043	0.2182 ± 0.0111	0.0313 ± 0.0296	0.3028 ± 0.0274
SelfRegulationSCP2	ACC	0.5163 ± 0.0051	0.5258 ± 0.0162	0.5811 ± 0.0143	0.5027 ± 0.0000	0.5521 ± 0.0096	0.5347 ± 0.0210	0.5053 ± 0.0069	0.5432 ± 0.0054	0.5811 ± 0.0402	0.5253 ± 0.0141
	F1	0.4913 ± 0.0063	0.5242 ± 0.0166	0.5657 ± 0.0301	0.7065 ± 0.0000	0.5448 ± 0.0077	0.5302 ± 0.0239	0.5039 ± 0.0073	0.5113 ± 0.0110	0.5676 ± 0.0372	0.5163 ± 0.0084
	NMI	0.0011 ± 0.0006	0.0027 ± 0.0027	0.0234 ± 0.0043	0.0000 ± 0.0000	0.0090 ± 0.0034	0.0049 ± 0.0048	0.0002 ± 0.0004	0.0083 ± 0.0020	0.0328 ± 0.0305	0.0034 ± 0.0044
	ARI	-0.0010 ± 0.0006	0.0011 ± 0.0036	0.0224 ± 0.0083	0.0000 ± 0.0000	0.0088 ± 0.0040	0.0041 ± 0.0065	-0.0023 ± 0.0005	0.0056 ± 0.0018	0.0313 ± 0.0296	0.0009 ± 0.0038
SpokenArabicDigits	ACC	0.4902 ± 0.0258	0.1000 ± 0.0000	0.2415 ± 0.0313	0.3869 ± 0.0543	0.4655 ± 0.1073	0.1889 ± 0.0452	0.3692 ± 0.0053	0.1000 ± 0.0000	0.3054 ± 0.0040	0.4420 ± 0.0257
	F1	0.4939 ± 0.0288	0.0182 ± 0.0000	0.2320 ± 0.0343	0.3303 ± 0.0362	0.4617 ± 0.1044	0.1196 ± 0.0519	0.3713 ± 0.0084	0.1819 ± 0.0000	0.3030 ± 0.0060	0.4460 ± 0.0206
	NMI	0.4740 ± 0.0129	0.0000 ± 0.0000	0.1566 ± 0.0474	0.4331 ± 0.0779	0.3946 ± 0.0887	0.1032 ± 0.0526	0.2691 ± 0.0144	0.0000 ± 0.0000	0.2431 ± 0.0002	0.4093 ± 0.0258
	ARI	0.3368 ± 0.0189	0.0000 ± 0.0000	0.0698 ± 0.0246	0.1351 ± 0.0703	0.2759 ± 0.0945	0.0529 ± 0.0279	0.1723 ± 0.0074	0.0000 ± 0.0000	0.1591 ± 0.0013	0.2643 ± 0.0280
StandWalkJump	ACC	0.4741 ± 0.0148	0.4593 ± 0.0502	0.6133 ± 0.0499	0.4444 ± 0.0406	0.5259 ± 0.0718	0.3333 ± 0.0000	0.4370 ± 0.0363	0.3926 ± 0.0181	0.5407 ± 0.0554	0.4296 ± 0.0646
	F1	0.4660 ± 0.0192	0.4545 ± 0.0527	0.6319 ± 0.0356	0.4183 ± 0.0631	0.5214 ± 0.0775	0.1667 ± 0.0000	0.4344 ± 0.0405	0.4802 ± 0.0242	0.5200 ± 0.0589	0.4069 ± 0.0675
	NMI	0.0997 ± 0.0235	0.1106 ± 0.0834	0.3446 ± 0.0358	0.1139 ± 0.0421	0.1219 ± 0.0596	0.0000 ± 0.0000	0.0866 ± 0.0489	0.0724 ± 0.0121	0.2150 ± 0.1475	0.0664 ± 0.0435
	ARI	0.0337 ± 0.0237	0.0184 ± 0.0621	0.1583 ± 0.0329	0.0137 ± 0.0393	0.0373 ± 0.0581	0.0000 ± 0.0000	-0.0145 ± 0.0314	-0.0146 ± 0.0145	0.1035 ± 0.1257	0.0083 ± 0.0492
UWaveGestureLibrary	ACC	0.5447 ± 0.0363	0.3073 ± 0.0291	0.3844 ± 0.0327	0.4614 ± 0.0993	0.3664 ± 0.0330	0.1250 ± 0.0000	0.3295 ± 0.0220	0.4559 ± 0.0421	0.5211 ± 0.0154	0.4550 ± 0.0410
	F1	0.5371 ± 0.0370	0.2952 ± 0.0280	0.3543 ± 0.0294	0.3472 ± 0.0630	0.3633 ± 0.0246	0.0278 ± 0.0000	0.3318 ± 0.0210	0.4536 ± 0.0487	0.5130 ± 0.0167	0.4412 ± 0.0434
	NMI	0.4631 ± 0.0409	0.1760 ± 0.0179	0.2606 ± 0.0468	0.3891 ± 0.0861	0.2394 ± 0.0115	0.0000 ± 0.0000	0.1658 ± 0.0284	0.3457 ± 0.0256	0.0022 ± 0.0026	0.3311 ± 0.0470
	ARI	0.3593 ± 0.0466	0.0901 ± 0.0148	0.1662 ± 0.0356	0.1692 ± 0.1171	0.1403 ± 0.0195	0.0000 ± 0.0000	0.0960 ± 0.0250	0.2268 ± 0.0340	0.0003 ± 0.0032	0.2016 ± 0.0428

jointly removing CCMA and MEVG lowers ARI on all 30 datasets. This pattern indicates that source-side admission and complementary view modeling address different aspects of information loss after pruning.

TABLEs V and VI reveal a similar division of labor between the two learning routes. Reconstruction maintains correspondence with the observed temporal structure, whereas contrastive refinement organizes the retained representation according to cluster geometry. Their relative importance remains dataset-dependent, but removing both routes produces the clearest degradation. On ArticularyWordRecognition, for instance, ARI decreases from 0.6029 to 0.4614 without reconstruction and to 0.5445 without contrastive refinement, while removing both reduces it to 0.0353. Comparable collapses appear on CharacterTrajectories and JapaneseVowels. These results show that the components do not contribute uniformly across datasets; instead, they provide complementary constraints whose coordination yields the most reliable overall behavior.

C. Additional Loss-Evolution Analysis

The main paper reports loss curves on four representative datasets. This subsection adds DuckDuckGeese, EthanolConcentration, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF to check whether the staged training process behaves similarly on more UEA cases. Fig. 2 reports the mean loss over five seeds, and the shaded bands denote standard deviation. The vertical dashed lines mark the transition from reconstruction-only training to dual-route training and later prototype-guided refinement.

TABLE III

ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART I. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
ArticularyWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
		✓	0.7155 ± 0.0137‡	0.7451 ± 0.0215‡	0.7824 ± 0.0132‡	0.5728 ± 0.0248‡
	✓		0.7739 ± 0.0437	0.7997 ± 0.0481	0.8255 ± 0.0189	0.6570 ± 0.0427
			0.6918 ± 0.0306‡	0.7336 ± 0.0245‡	0.7709 ± 0.0187‡	0.5517 ± 0.0369‡
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
		✓	0.4867 ± 0.0163‡	0.4331 ± 0.0116‡	0.1050 ± 0.0047‡	0.0404 ± 0.0051‡
	✓		0.4800 ± 0.0400‡	0.4681 ± 0.0854‡	0.1365 ± 0.0448	0.0664 ± 0.0421
			0.4600 ± 0.0327‡	0.4024 ± 0.0485‡	0.1000 ± 0.0087‡	0.0346 ± 0.0071‡
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
		✓	0.9825 ± 0.0127	0.9824 ± 0.0129	0.9497 ± 0.0366	0.9533 ± 0.0338
	✓		0.8900 ± 0.1317‡	0.9203 ± 0.0746‡	0.8924 ± 0.0754‡	0.8361 ± 0.1360‡
			0.8700 ± 0.0100‡	0.9098 ± 0.0102‡	0.9138 ± 0.0287‡	0.8202 ± 0.0263‡
Character Trajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
		✓	0.7824 ± 0.0547‡	0.8177 ± 0.0436	0.8416 ± 0.0367‡	0.7078 ± 0.0644‡
	✓		0.7371 ± 0.0227‡	0.7610 ± 0.0346‡	0.8038 ± 0.0198‡	0.6417 ± 0.0303‡
			0.7369 ± 0.0590‡	0.7699 ± 0.0708‡	0.8138 ± 0.0468‡	0.6556 ± 0.0750‡
Cricket	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
		✓	0.8489 ± 0.0349	0.8739 ± 0.0420	0.8737 ± 0.0444	0.7589 ± 0.0659
	✓		0.7811 ± 0.0658‡	0.8191 ± 0.0588‡	0.8346 ± 0.0364‡	0.6817 ± 0.0702‡
			0.8078 ± 0.0114‡	0.8292 ± 0.0241‡	0.8376 ± 0.0130‡	0.6937 ± 0.0195‡
DuckDuckGeese	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
		✓	0.3700 ± 0.0179‡	0.3600 ± 0.0253‡	0.1143 ± 0.0234‡	0.0526 ± 0.0108‡
	✓		0.3280 ± 0.0421‡	0.2966 ± 0.0425‡	0.0869 ± 0.0246‡	0.0196 ± 0.0205‡
			0.3680 ± 0.0194‡	0.3449 ± 0.0235‡	0.1229 ± 0.0145‡	0.0306 ± 0.0112‡
ERing	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
		✓	0.6867 ± 0.0669‡	0.7180 ± 0.0341‡	0.5238 ± 0.0426‡	0.5008 ± 0.0669‡
	✓		0.7300 ± 0.0485‡	0.7113 ± 0.0475‡	0.5198 ± 0.0400‡	0.5026 ± 0.0569‡
			0.7173 ± 0.0259‡	0.7082 ± 0.0253‡	0.5112 ± 0.0390‡	0.5033 ± 0.0415‡
EigenWorms	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
		✓	0.4131 ± 0.0101‡	0.3546 ± 0.0347‡	0.1196 ± 0.0211‡	0.1907 ± 0.0156
	✓		0.4125 ± 0.0451‡	0.3577 ± 0.0359‡	0.1329 ± 0.0256‡	0.1469 ± 0.0344‡
			0.3822 ± 0.0462‡	0.3397 ± 0.0392‡	0.0713 ± 0.0600‡	0.1368 ± 0.0716‡
Epilepsy	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
		✓	0.4640 ± 0.0291‡	0.4448 ± 0.0300‡	0.2217 ± 0.0332‡	0.1511 ± 0.0259‡
	✓		0.4420 ± 0.0368‡	0.4505 ± 0.0382‡	0.2330 ± 0.0451‡	0.1551 ± 0.0279‡
			0.3829 ± 0.0238‡	0.4047 ± 0.0243‡	0.2118 ± 0.0298‡	0.1637 ± 0.0212‡
Ethanol Concentration	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
		✓	0.3015 ± 0.0099‡	0.2577 ± 0.0082‡	0.0114 ± 0.0033‡	0.0049 ± 0.0030‡
	✓		0.3058 ± 0.0183‡	0.2581 ± 0.0252‡	0.0184 ± 0.0066‡	0.0104 ± 0.0062‡
			0.2950 ± 0.0092‡	0.2397 ± 0.0108‡	0.0124 ± 0.0061‡	0.0045 ± 0.0039‡
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
		✓	0.5051 ± 0.0028‡	0.5050 ± 0.0028‡	0.0001 ± 0.0001‡	0.0000 ± 0.0001‡
	✓		0.5101 ± 0.0023‡	0.5092 ± 0.0027‡	0.0003 ± 0.0001‡	0.0003 ± 0.0002‡
			0.5063 ± 0.0076‡	0.5061 ± 0.0077‡	0.0003 ± 0.0005‡	0.0003 ± 0.0007‡
FingerMovements	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
		✓	0.5168 ± 0.0101‡	0.5123 ± 0.0056‡	0.0013 ± 0.0015‡	0.0008 ± 0.0017
	✓		0.5168 ± 0.0128‡	0.5125 ± 0.0160‡	0.0013 ± 0.0014‡	0.0005 ± 0.0019‡
			0.5130 ± 0.0074‡	0.5127 ± 0.0073‡	0.0006 ± 0.0007‡	0.0003 ± 0.0002‡
HandMovement Direction	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
		✓	0.3051 ± 0.0154‡	0.3027 ± 0.0162‡	0.0139 ± 0.0036‡	0.0008 ± 0.0029‡
	✓		0.3179 ± 0.0249‡	0.3148 ± 0.0246‡	0.0201 ± 0.0119‡	0.0049 ± 0.0111‡
			0.3060 ± 0.0126‡	0.3034 ± 0.0116‡	0.0144 ± 0.0029‡	0.0011 ± 0.0025‡
Handwriting	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
		✓	0.2102 ± 0.0132‡	0.2103 ± 0.0137‡	0.3030 ± 0.0188‡	0.0770 ± 0.0111‡
	✓		0.2444 ± 0.0113	0.2460 ± 0.0156	0.3503 ± 0.0160	0.1050 ± 0.0109
			0.2224 ± 0.0130‡	0.2234 ± 0.0113‡	0.3212 ± 0.0209‡	0.0852 ± 0.0100‡
Heartbeat	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
		✓	0.5826 ± 0.0369‡	0.5002 ± 0.0271‡	0.0438 ± 0.0108	0.0952 ± 0.0277
	✓		0.5841 ± 0.0410‡	0.5474 ± 0.0747‡	0.0328 ± 0.0106‡	0.0620 ± 0.0591‡
			0.5066 ± 0.0091‡	0.4463 ± 0.0947‡	0.0365 ± 0.0108‡	0.0346 ± 0.0516‡

TABLE IV

ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART II. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
		✓	0.1351 ± 0.0015	0.1206 ± 0.0036	0.0147 ± 0.0008‡	0.0064 ± 0.0001
	✓		0.1304 ± 0.0032‡	0.1115 ± 0.0076‡	0.0132 ± 0.0011‡	0.0048 ± 0.0013‡
			0.1323 ± 0.0039‡	0.1099 ± 0.0069‡	0.0139 ± 0.0017‡	0.0050 ± 0.0011‡
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
		✓	0.8162 ± 0.0393‡	0.8557 ± 0.0416‡	0.8044 ± 0.0126‡	0.7321 ± 0.0429‡
	✓		0.8050 ± 0.0347‡	0.8440 ± 0.0294‡	0.8220 ± 0.0239	0.7847 ± 0.0385
			0.7800 ± 0.0306‡	0.7802 ± 0.0270‡	0.8140 ± 0.0272‡	0.7648 ± 0.0553‡
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
		✓	0.2018 ± 0.0106‡	0.1753 ± 0.0156‡	0.1375 ± 0.0125‡	0.0361 ± 0.0074‡
	✓		0.1996 ± 0.0131‡	0.1792 ± 0.0080‡	0.1358 ± 0.0102‡	0.0267 ± 0.0093‡
			0.1748 ± 0.0134‡	0.1624 ± 0.0031‡	0.1053 ± 0.0062‡	0.0189 ± 0.0052‡
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
		✓	0.2839 ± 0.0032‡	0.2979 ± 0.0079‡	0.3439 ± 0.0107‡	0.1012 ± 0.0057‡
	✓		0.3272 ± 0.0278	0.3460 ± 0.0497	0.4212 ± 0.0300	0.1600 ± 0.0271
			0.2883 ± 0.0069‡	0.2959 ± 0.0148‡	0.3481 ± 0.0125‡	0.1084 ± 0.0080‡
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
		✓	0.5260 ± 0.0178‡	0.5196 ± 0.0260‡	0.0053 ± 0.0041	0.0041 ± 0.0050
	✓		0.5307 ± 0.0224‡	0.5249 ± 0.0284‡	0.0042 ± 0.0048	0.0032 ± 0.0066
			0.5238 ± 0.0115‡	0.4965 ± 0.0034‡	0.0035 ± 0.0038‡	0.0007 ± 0.0027‡
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
		✓	0.4533 ± 0.0457‡	0.4550 ± 0.0494‡	0.4481 ± 0.0457‡	0.2797 ± 0.0460‡
	✓		0.4528 ± 0.0511‡	0.4554 ± 0.0583‡	0.4444 ± 0.0859‡	0.2844 ± 0.0753‡
			0.4750 ± 0.0072‡	0.4097 ± 0.0059‡	0.4053 ± 0.0118‡	0.3082 ± 0.0133‡
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
		✓	0.2768 ± 0.0192‡	0.2752 ± 0.0175‡	0.2469 ± 0.0372‡	0.1170 ± 0.0271‡
	✓		0.3159 ± 0.0708‡	0.3722 ± 0.0819‡	0.3329 ± 0.0928‡	0.1465 ± 0.0908‡
			0.2550 ± 0.0164‡	0.2553 ± 0.0131‡	0.2111 ± 0.0293‡	0.0900 ± 0.0244‡
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
		✓	0.5957 ± 0.0208‡	0.5949 ± 0.0207‡	0.5736 ± 0.0164‡	0.4065 ± 0.0264‡
	✓		0.6614 ± 0.0378‡	0.6457 ± 0.0436‡	0.6188 ± 0.0319‡	0.4953 ± 0.0431‡
			0.5954 ± 0.0262‡	0.5955 ± 0.0250‡	0.5702 ± 0.0197‡	0.4034 ± 0.0284‡
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
		✓	0.1042 ± 0.0026‡	0.1061 ± 0.0013‡	0.1636 ± 0.0047‡	0.0296 ± 0.0047‡
	✓		0.1044 ± 0.0018‡	0.1029 ± 0.0008‡	0.1576 ± 0.0107‡	0.0277 ± 0.0029‡
			0.1036 ± 0.0016‡	0.1027 ± 0.0021‡	0.1568 ± 0.0039‡	0.0289 ± 0.0014‡
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
		✓	0.5330 ± 0.0454‡	0.5263 ± 0.0497‡	0.3037 ± 0.0637‡	0.2443 ± 0.0751‡
	✓		0.5340 ± 0.0979‡	0.5410 ± 0.0966‡	0.3100 ± 0.1019‡	0.2449 ± 0.1091‡
			0.4026 ± 0.0949‡	0.4012 ± 0.0964‡	0.2912 ± 0.1121‡	0.1290 ± 0.1141‡
SelfRegulation SCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
		✓	0.7647 ± 0.0254‡	0.7635 ± 0.0258‡	0.2224 ± 0.0410‡	0.2816 ± 0.0510‡
	✓		0.7740 ± 0.0170‡	0.7730 ± 0.0160‡	0.2387 ± 0.0412	0.3002 ± 0.0376‡
			0.7647 ± 0.0103‡	0.7644 ± 0.0099‡	0.2150 ± 0.0213‡	0.2794 ± 0.0224‡
SelfRegulation SCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
		✓	0.5279 ± 0.0107	0.5270 ± 0.0093	0.0026 ± 0.0022‡	0.0010 ± 0.0029‡
	✓		0.5147 ± 0.0106‡	0.5141 ± 0.0105‡	0.0010 ± 0.0013‡	0.0010 ± 0.0017‡
			0.5142 ± 0.0032‡	0.5141 ± 0.0033‡	0.0006 ± 0.0003‡	0.0008 ± 0.0004‡
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
		✓	0.3712 ± 0.0196‡	0.3778 ± 0.0219‡	0.3407 ± 0.0153‡	0.1844 ± 0.0124‡
	✓		0.4177 ± 0.0191‡	0.4002 ± 0.0231‡	0.3457 ± 0.0383‡	0.1802 ± 0.0395‡
			0.3637 ± 0.0203‡	0.3691 ± 0.0204‡	0.3226 ± 0.0119‡	0.1621 ± 0.0086‡
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
		✓	0.3800 ± 0.0277‡	0.3549 ± 0.0372‡	0.0408 ± 0.0276‡	0.0133 ± 0.0277‡
	✓		0.3822 ± 0.0502‡	0.3544 ± 0.0506‡	0.0417 ± 0.0306‡	0.0135 ± 0.0349‡
			0.3148 ± 0.0544‡	0.2852 ± 0.0626‡	0.0347 ± 0.0419‡	0.0101 ± 0.0161‡
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
		✓	0.3373 ± 0.0086‡	0.3156 ± 0.0090‡	0.1918 ± 0.0200‡	0.0954 ± 0.0114‡
	✓		0.3945 ± 0.0411‡	0.3881 ± 0.0581‡	0.2768 ± 0.0378‡	0.1487 ± 0.0261‡
			0.3127 ± 0.0232‡	0.2967 ± 0.0177‡	0.1782 ± 0.0133‡	0.0860 ± 0.0156‡

In the first stage, the loss decreases on all six datasets, indicating that admitted patches can support sequence recovery after part of the input is removed. After $\mathcal{L} * \mathrm{orth}$ and $\mathcal{L} * \mathrm{pa}$ enter training, several curves rise or fluctuate for a short period because the objective no longer measures reconstruction alone. The model begins to adjust view diversity and prototype alignment, so a temporary increase is expected before the route-level terms settle. In later epochs, DuckDuckGeese, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF show smoother refinement. EthanolConcentration has a wider variance band near the transition, but the curve also becomes controlled afterward. The additional curves support the staged training order by showing that route-level refinement does not cause persistent divergence.

D. Mask-Rate Scalability Analysis

The main paper reports runtime and GPU memory when N, T, and D increase. This subsection checks the source of the runtime saving more directly. If admission reduces encoder-side computation, then increasing the mask rate should lower normalized execution time because fewer patches enter attention. Fig. 3 tests this relation by varying N, T, and D separately.

Given patch length ℓ_{p}, each sequence is divided into $m = \lceil T/\ell_{\mathrm{p}} \rceil$ patches. Under retained ratio r_{a}, CCMA sends $m_{\mathrm{a}} = \lceil r_{\mathrm{a}} m \rceil$ patches to the encoder. The mask rate is defined as

$$r_{\mathrm{m}} = 1 - \frac{m_{\mathrm{a}}}{m}. \tag{1}$$

The experiment increases r_{m} from 0% to 90% by decreasing r_{a}, and execution time is normalized by the zero-mask cost for comparison across input scales.

The curves in Fig. 3 show that runtime generally drops as r_{m} increases. The average reduction reaches 78.2% when varying N, 86.3% when varying T, and 78.8% when varying D. The largest decrease appears in the sequence-length test, which is consistent with the complexity analysis because attention cost depends on the retained patch number. The same trend under larger N and D indicates that the saving is not tied to a single scaling factor. Overall, the mask-rate curves support the role of

TABLE V

ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART I. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
ArticularyWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
		✓	0.6035 ± 0.0317‡	0.6212 ± 0.0261‡	0.7016 ± 0.0278‡	0.4614 ± 0.0399‡
	✓		0.6247 ± 0.0562‡	0.6514 ± 0.0558‡	0.6747 ± 0.0426‡	0.5445 ± 0.0793‡
			0.1419 ± 0.0485‡	0.2366 ± 0.0323‡	0.2485 ± 0.0639‡	0.0353 ± 0.0299‡
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
		✓	0.4933 ± 0.0249	0.4763 ± 0.0291	0.0833 ± 0.0242‡	0.0169 ± 0.0203‡
	✓		0.4733 ± 0.0389‡	0.4520 ± 0.0542‡	0.0765 ± 0.0312‡	0.0190 ± 0.0333‡
			0.4600 ± 0.0389‡	0.4518 ± 0.0598‡	0.0937 ± 0.0356‡	0.0212 ± 0.0325‡
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
		✓	0.9050 ± 0.1226‡	0.9319 ± 0.0784‡	0.9161 ± 0.0911‡	0.8633 ± 0.1575‡
	✓		0.6600 ± 0.0366‡	0.8105 ± 0.0355‡	0.7754 ± 0.0499‡	0.6063 ± 0.0482‡
			0.5100 ± 0.0289‡	0.6355 ± 0.0519‡	0.5518 ± 0.0478‡	0.3927 ± 0.0402‡
Character Trajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
		✓	0.6714 ± 0.0832‡	0.7199 ± 0.0942‡	0.7443 ± 0.0645‡	0.5830 ± 0.0923‡
	✓		0.7389 ± 0.0708‡	0.7674 ± 0.0674‡	0.8189 ± 0.0503‡	0.6750 ± 0.0821‡
			0.0651 ± 0.0000‡	0.1222 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
Cricket	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
		✓	0.7900 ± 0.1095‡	0.8001 ± 0.1052‡	0.8279 ± 0.0699‡	0.6912 ± 0.1158‡
	✓		0.7511 ± 0.0708‡	0.7828 ± 0.0339‡	0.7682 ± 0.0476‡	0.5964 ± 0.0836‡
			0.1000 ± 0.0333‡	0.2018 ± 0.0960‡	0.0721 ± 0.1442‡	0.0121 ± 0.0241‡
DuckDuckGeese	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
		✓	0.4740 ± 0.0546	0.4690 ± 0.0553	0.2439 ± 0.0457	0.1618 ± 0.0523
	✓		0.3880 ± 0.0306‡	0.3682 ± 0.0377‡	0.1562 ± 0.0375‡	0.0621 ± 0.0248‡
			0.2000 ± 0.0000‡	0.3333 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
ERing	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
		✓	0.6533 ± 0.0419‡	0.6553 ± 0.0421‡	0.5735 ± 0.0240‡	0.4854 ± 0.0353‡
	✓		0.6020 ± 0.0937‡	0.6006 ± 0.0951‡	0.5408 ± 0.0702‡	0.4688 ± 0.1070‡
			0.1667 ± 0.0000‡	0.2857 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
EigenWorms	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
		✓	0.4556 ± 0.0523	0.4150 ± 0.0632	0.2774 ± 0.0369	0.2479 ± 0.0381
	✓		0.3382 ± 0.0317‡	0.3009 ± 0.0439‡	0.0843 ± 0.0295‡	0.0512 ± 0.0158‡
			0.2247 ± 0.0000‡	0.2962 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
Epilepsy	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
		✓	0.5716 ± 0.0429	0.5746 ± 0.0356	0.3998 ± 0.0509	0.3182 ± 0.0499
	✓		0.4320 ± 0.0292‡	0.4259 ± 0.0276‡	0.2047 ± 0.0333‡	0.1442 ± 0.0279‡
			0.4291 ± 0.0332‡	0.4217 ± 0.0355‡	0.1670 ± 0.0517‡	0.1224 ± 0.0370‡
Ethanol Concentration	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
		✓	0.2916 ± 0.0080‡	0.2541 ± 0.0078‡	0.0129 ± 0.0041‡	0.0032 ± 0.0014‡
	✓		0.3015 ± 0.0143‡	0.2759 ± 0.0252‡	0.0108 ± 0.0187‡	0.0044 ± 0.0006‡
			0.1179 ± 0.0130‡	0.1277 ± 0.0507‡	0.0042 ± 0.0022‡	0.0038 ± 0.0027‡
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
		✓	0.5090 ± 0.0055‡	0.5086 ± 0.0056‡	0.0003 ± 0.0003‡	0.0003 ± 0.0004‡
	✓		0.5000 ± 0.0000‡	0.4667 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.5000 ± 0.0000‡	0.4667 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
FingerMovements	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
		✓	0.5135 ± 0.0097‡	0.5114 ± 0.0111‡	0.0008 ± 0.0009‡	0.0013 ± 0.0012
	✓		0.5031 ± 0.0102‡	0.5088 ± 0.0102‡	0.0009 ± 0.0002‡	0.0002 ± 0.0016‡
			0.5000 ± 0.0000‡	0.4667 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
HandMovement Direction	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
		✓	0.3077 ± 0.0162‡	0.3061 ± 0.0171‡	0.0148 ± 0.0062‡	0.0005 ± 0.0053‡
	✓		0.3111 ± 0.0208‡	0.3077 ± 0.0221‡	0.0215 ± 0.0075	0.0053 ± 0.0072‡
			0.3068 ± 0.0068‡	0.2901 ± 0.0134‡	0.0201 ± 0.0043‡	0.0021 ± 0.0020‡
Handwriting	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
		✓	0.1706 ± 0.0061‡	0.1741 ± 0.0083‡	0.2227 ± 0.0045‡	0.0385 ± 0.0019‡
	✓		0.2054 ± 0.0071‡	0.2071 ± 0.0059‡	0.2809 ± 0.0091‡	0.0663 ± 0.0054‡
			0.1428 ± 0.0060‡	0.1408 ± 0.0057‡	0.1813 ± 0.0062‡	0.0217 ± 0.0024‡
Heartbeat	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
		✓	0.5628 ± 0.0376‡	0.5053 ± 0.0802‡	0.0459 ± 0.0155	0.0065 ± 0.0412‡
	✓		0.5002 ± 0.0176‡	0.5537 ± 0.1522‡	0.0170 ± 0.0068‡	0.0161 ± 0.0034‡
			0.5213 ± 0.0000‡	0.4381 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡

TABLE VI

ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART II. RED "‡" MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
		✓	0.1206 ± 0.0063‡	0.1104 ± 0.0111‡	0.0053 ± 0.0027‡	0.0025 ± 0.0014‡
	✓		0.1000 ± 0.0000‡	0.1018 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.1000 ± 0.0000‡	0.1018 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
		✓	0.8059 ± 0.0310‡	0.8042 ± 0.0270‡	0.7824 ± 0.0315‡	0.7327 ± 0.0522‡
	✓		0.6441 ± 0.1450‡	0.6632 ± 0.1467‡	0.5833 ± 0.1473‡	0.4328 ± 0.1564‡
			0.1844 ± 0.0000‡	0.3113 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
		✓	0.3362 ± 0.0017	0.2228 ± 0.0094	0.1699 ± 0.0142	0.0714 ± 0.0062
	✓		0.1969 ± 0.0139‡	0.1776 ± 0.0102‡	0.1427 ± 0.0084‡	0.0537 ± 0.0075‡
			0.1155 ± 0.0000‡	0.0797 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
		✓	0.3000 ± 0.0283‡	0.3072 ± 0.0231‡	0.3518 ± 0.0285‡	0.1016 ± 0.0259‡
	✓		0.3061 ± 0.0210‡	0.3003 ± 0.0315‡	0.3402 ± 0.0261‡	0.1022 ± 0.0231‡
			0.2644 ± 0.0235‡	0.2673 ± 0.0222‡	0.3222 ± 0.0456‡	0.0927 ± 0.0307‡
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
		✓	0.5079 ± 0.0044‡	0.4726 ± 0.0036‡	0.0004 ± 0.0004‡	0.0016 ± 0.0004‡
	✓		0.5270 ± 0.0074‡	0.5048 ± 0.0179‡	0.0034 ± 0.0023‡	0.0009 ± 0.0017‡
			0.1000 ± 0.0313‡	0.2218 ± 0.0950‡	0.0631 ± 0.0442‡	0.0221 ± 0.0235‡
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
		✓	0.4972 ± 0.0478	0.4940 ± 0.0434	0.4796 ± 0.0594	0.3245 ± 0.0508
	✓		0.4328 ± 0.0428‡	0.4242 ± 0.0431‡	0.3578 ± 0.0955‡	0.2377 ± 0.0650‡
			0.1967 ± 0.0600‡	0.3526 ± 0.1337‡	0.0770 ± 0.1539‡	0.0221 ± 0.0441‡
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
		✓	0.2368 ± 0.0216‡	0.2476 ± 0.0355‡	0.1705 ± 0.0511‡	0.0609 ± 0.0218‡
	✓		0.2700 ± 0.0273‡	0.2785 ± 0.0369‡	0.1930 ± 0.0781‡	0.0771 ± 0.0412‡
			0.1477 ± 0.0000‡	0.2574 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
		✓	0.7139 ± 0.0446	0.7030 ± 0.0448	0.6472 ± 0.0277	0.5420 ± 0.0420
	✓		0.1041 ± 0.0000‡	0.1885 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.1041 ± 0.0000‡	0.1885 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
		✓	0.0985 ± 0.0017‡	0.0967 ± 0.0012‡	0.1443 ± 0.0051‡	0.0243 ± 0.0012‡
	✓		0.0256 ± 0.0000‡	0.0500 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.0256 ± 0.0000‡	0.0500 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
		✓	0.6389 ± 0.0424	0.6436 ± 0.0345	0.4214 ± 0.0229	0.3526 ± 0.0483
	✓		0.4488 ± 0.0172‡	0.4449 ± 0.0185‡	0.1929 ± 0.0490‡	0.1187 ± 0.0352‡
			0.4244 ± 0.0367‡	0.3947 ± 0.0725‡	0.1550 ± 0.0234‡	0.1061 ± 0.0245‡
Self RegulationSCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
		✓	0.7458 ± 0.0132‡	0.7421 ± 0.0156‡	0.2011 ± 0.0274‡	0.2411 ± 0.0261‡
	✓		0.7337 ± 0.0714‡	0.7246 ± 0.0859‡	0.1956 ± 0.0777‡	0.2376 ± 0.1041‡
			0.7480 ± 0.0175‡	0.7477 ± 0.0176‡	0.1878 ± 0.0261‡	0.2458 ± 0.0341‡
Self RegulationSCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
		✓	0.5347 ± 0.0139	0.5325 ± 0.0152	0.0041 ± 0.0021	0.0030 ± 0.0029
	✓		0.5158 ± 0.0173‡	0.5106 ± 0.0157‡	0.0017 ± 0.0031‡	0.0003 ± 0.0040‡
			0.5200 ± 0.0061‡	0.5191 ± 0.0059	0.0013 ± 0.0007‡	0.0009 ± 0.0009‡
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
		✓	0.3631 ± 0.0512‡	0.3642 ± 0.0468‡	0.3212 ± 0.0486‡	0.1523 ± 0.0453‡
	✓		0.1000 ± 0.0000‡	0.1819 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
			0.1000 ± 0.0000‡	0.1819 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
		✓	0.3593 ± 0.0378‡	0.3537 ± 0.0361‡	0.0330 ± 0.0194‡	0.0124 ± 0.0251‡
	✓		0.3519 ± 0.0363‡	0.3355 ± 0.0497‡	0.0343 ± 0.0236‡	0.0138 ± 0.0199‡
			0.3296 ± 0.0181‡	0.2740 ± 0.0214‡	0.0124 ± 0.0506‡	0.0016 ± 0.0243‡
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
		✓	0.4023 ± 0.0342‡	0.3992 ± 0.0359‡	0.2887 ± 0.0371‡	0.1684 ± 0.0280‡
	✓		0.6318 ± 0.0456	0.6520 ± 0.0381	0.5779 ± 0.0160	0.4497 ± 0.0166
			0.1250 ± 0.0000‡	0.2222 ± 0.0000‡	0.0000 ± 0.0000‡	0.0000 ± 0.0000‡

CCMA in reducing encoder-side cost through fewer admitted patches.

E. Noise and Embedding Robustness Diagnostics

Fig. 4 supplements Q4 with two controlled robustness tests on AtrialFibrillation and Libras. The noise test adds Gaussian perturbation to the input sequence and examines whether clustering cues remain recognizable when local observations become less reliable. The embedding test reduces the latent dimension from 64 to 4 and examines whether the learned partition remains usable after representation compression. The two settings target different failure cases after pruning. A desirable retained representation should avoid overreacting to noisy patches and should keep cluster-related structure without relying on a large embedding space.

In Fig. 4(a), the F1-score curves show that SIGMA changes more smoothly as the noise level increases. On AtrialFibrillation, SIGMA stays above the compared baselines across the tested perturbation range. On Libras, several methods fluctuate after noise is injected, whereas SIGMA shows milder variation. The trend suggests that confidence-calibrated admission reduces the influence of corrupted or routine patches before encoder-side interaction, and the retained endogenous views still provide enough temporal context for clustering.

Fig. 4(b) evaluates embedding-dimension robustness by compressing the latent space. As the dimension decreases, SIGMA remains competitive on both datasets and keeps a steadier trajectory across the tested dimensions. EMTC is competitive on Libras at the smallest dimension, but its behavior is less consistent across the full range. The dimension test suggests that SIGMA does not obtain clustering quality from an oversized latent representation. Taken together, the two diagnostics show that SIGMA preserves usable clustering cues under input noise and embedding compression.

F. Semantic Preservation Evaluation

To directly assess whether the compact representations produced by SIGMA retain the structural information needed for

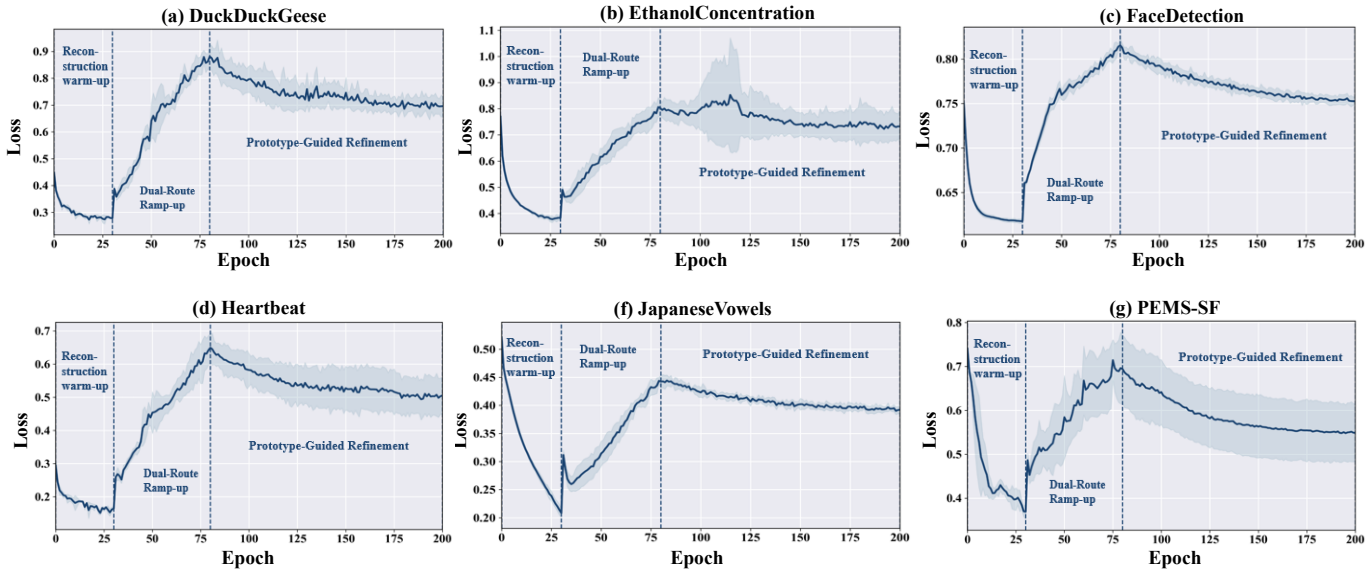

Fig. 2. Additional loss-evolution curves during SIGMA training on six UEA datasets. Each curve reports the mean loss across five random seeds, and the shaded band denotes standard deviation. Panels (a) to (f) correspond to DuckDuckGeese, EthanolConcentration, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF, respectively. Vertical dashed lines mark the reconstruction-only, dual-route, and prototype-guided refinement stages.

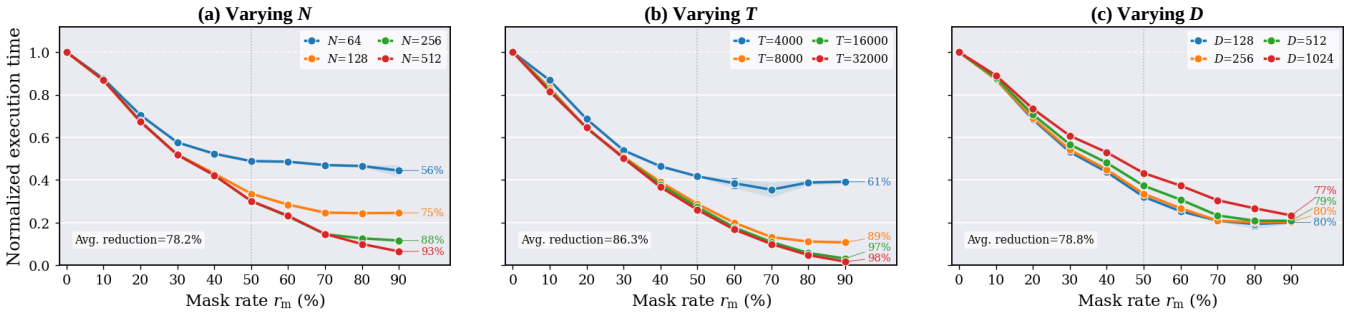

Fig. 3. Scalability under increasing mask rates. The three panels vary N, T, and D, respectively. For each setting, execution time is normalized by the cost at zero mask rate. Shaded bands denote standard deviation, and endpoint annotations report the reduction at 90% mask rate.

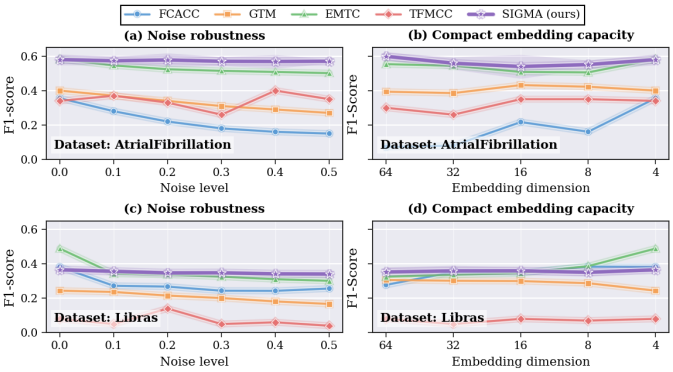

Fig. 4. **Robustness and compactness diagnostics.** F1-score on AtrialFibrillation and Libras under (a) increasing Gaussian noise and (b) reduced embedding dimensions from 64 to 4.

reliable clustering, we introduce Centered Kernel Alignment (CKA) [10], a representation-similarity measure that compares the relational structure of learned representation spaces without relying on reconstruction loss or clustering labels. Higher CKA indicates stronger preservation of representation-level structure, providing a complementary criterion to the reconstruction-derived SF used in the main paper. TABLE VII compares SIGMA with FCACC, TFMCC, and EMTC on seven representative datasets under the same evaluation protocol. SIGMA achieves the highest CKA on all seven datasets, with an average score of 0.5163, compared with 0.3758 for FCACC, 0.3706 for TFMCC, and 0.2049 for EMTC. The advantage is particularly clear on ERing, JapaneseVowels, and SelfRegulationSCP1, where SIGMA reaches 0.6478, 0.8197, and 0.6633, respectively. These results provide loss-independent evidence that the compact representations produced after source-side admission retain substantially more

TABLE VII
CKA-BASED SEMANTIC PRESERVATION COMPARISON ON SEVEN REPRESENTATIVE DATASETS. HIGHER CKA INDICATES STRONGER REPRESENTATION-LEVEL STRUCTURAL CONSISTENCY, AND **BOLD** DENOTES THE BEST MEAN RESULT.

Datasets	FCACC	TFMCC	EMTC	SIGMA (ours)
ArticularyWordRecognition	0.3591 ± 0.0199	0.3366 ± 0.0092	0.1254 ± 0.0429	**0.4799 ± 0.0379**
BasicMotions	0.2998 ± 0.0030	0.2789 ± 0.0560	0.2028 ± 0.0210	**0.3329 ± 0.0036**
CharacterTrajectories	0.3134 ± 0.0097	0.3346 ± 0.0216	0.1772 ± 0.1083	**0.3887 ± 0.0504**
ERing	0.4369 ± 0.0029	0.3835 ± 0.0132	0.4001 ± 0.1442	**0.6478 ± 0.0198**
JapaneseVowels	0.6933 ± 0.0219	0.6136 ± 0.0187	0.1106 ± 0.0311	**0.8197 ± 0.0141**
MotorImagery	0.2535 ± 0.0721	0.2037 ± 0.0072	0.1803 ± 0.0092	**0.2820 ± 0.0278**
SelfRegulationSCP1	0.2745 ± 0.0076	0.4431 ± 0.1270	0.2377 ± 0.0053	**0.6633 ± 0.0432**

representation-level structure, complementing SF and strengthening the empirical support for the semantic-preservation side of the prune-but-preserve design.

G. Fine-Grained Mechanism Ablation

To further isolate the mechanisms underlying SIGMA beyond the module- and route-level ablations, we conduct a finer-grained study on the same ten representative datasets by modifying one internal design at a time. For CCMA, we separately remove hard admission, adaptive ranking, and confidence calibration; for MEVG, we remove multi-view modeling or replace independent encoders with shared ones. All remaining training settings are kept unchanged. As shown in TABLE VIII, the complete SIGMA obtains the highest average performance over the ten datasets on all four metrics, reaching 0.5274 ACC, 0.5156 F1, 0.3361 NMI, and 0.2754 ARI. The largest ARI reductions arise from removing adaptive ranking and confidence calibration, which decrease the average ARI by 22.9% and 20.6%, respectively, indicating that dynamically identifying informative patches and retaining their confidence information are the main mechanisms within CCMA. Removing hard admission further reduces ARI by 11.3%, supporting the benefit of controlling patch participation before contextual interaction. Within MEVG, removing multi-view modeling or independent encoders decreases average ARI by 9.6% and 9.6%, respectively, showing that view diversity and view-specific encoding provide complementary gains. These controlled variants clarify that the observed improvement does not arise uniformly from an accumulation of components: adaptive ranking and confidence calibration contribute most strongly, while hard admission, multi-view modeling, and encoder specialization provide additional coordinated improvements.

IV. SUPPLEMENTARY RESULTS

This section provides the full experimental record behind the condensed evaluation in the main paper under the same unsupervised protocol defined in Section II. Complete benchmark tables and Bonferroni-Dunn ranks extend Q1 to all 30 UEA datasets. Architecture, route-internal, and fine-grained mechanism ablations expand Q2 by examining SIGMA from module, objective, and internal-mechanism levels. The additional CKA-based evaluation further examines semantic preservation by comparing representation-level structural consistency with representative clustering methods. Larger-scale and mask-rate tests examine Q3, while additional robustness diagnostics complement Q4 under noise perturbation and embedding compression. Together, these supplementary results provide more detailed evidence for clustering effectiveness, mechanism attribution, semantic preservation, computational efficiency, and robustness without introducing a separate evaluation setting.

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TABLE VIII

FINE-GRAINED MECHANISM ABLATION ON TEN REPRESENTATIVE DATASETS. EACH VARIANT REMOVES OR REPLACES ONE INTERNAL MECHANISM WHILE KEEPING THE REMAINING SETTINGS UNCHANGED. THE BEST MEAN RESULT FOR EACH DATASET AND METRIC IS HIGHLIGHTED IN BOLD.

Dataset	Metric	w/o Hard Admission	w/o Adaptive Ranking	w/o Confidence Calibration	w/o Multi-View	w/o Independent Encoders	SIGMA (full)
ArticularyWordRecognition	ACC	0.7089 ± 0.0207	0.7075 ± 0.0355	0.6623 ± 0.0514	0.6880 ± 0.0399	0.6838 ± 0.0177	**0.7603 ± 0.0232**
	F1	0.7509 ± 0.0169	0.7592 ± 0.0332	0.6796 ± 0.0427	0.7071 ± 0.0410	0.7127 ± 0.0134	**0.7929 ± 0.0392**
	NMI	0.7769 ± 0.0164	0.7883 ± 0.0253	0.7305 ± 0.0317	0.7532 ± 0.0312	0.7524 ± 0.0148	**0.8276 ± 0.0165**
	ARI	0.5629 ± 0.0255	0.5724 ± 0.0442	0.5058 ± 0.0511	0.5336 ± 0.0533	0.5373 ± 0.0203	**0.6466 ± 0.0257**
BasicMotions	ACC	0.9825 ± 0.0127	0.6375 ± 0.0079	0.9750 ± 0.0224	**0.9875 ± 0.0112**	0.9850 ± 0.0094	**0.9875 ± 0.0079**
	F1	0.9824 ± 0.0129	0.7890 ± 0.0070	0.9748 ± 0.0227	**0.9875 ± 0.0112**	0.9850 ± 0.0094	**0.9875 ± 0.0079**
	NMI	0.9496 ± 0.0366	0.7530 ± 0.0186	0.9450 ± 0.0403	0.9665 ± 0.0097	0.9593 ± 0.0246	**0.9665 ± 0.0191**
	ARI	0.9531 ± 0.0339	0.5903 ± 0.0160	0.9369 ± 0.0537	0.9666 ± 0.0193	0.9600 ± 0.0247	**0.9666 ± 0.0208**
DuckDuckGeese	ACC	0.3880 ± 0.0319	0.3840 ± 0.0441	0.3860 ± 0.0233	0.3240 ± 0.0441	0.3620 ± 0.0549	**0.4020 ± 0.0325**
	F1	0.3680 ± 0.0372	0.3689 ± 0.0408	0.3726 ± 0.0284	0.3007 ± 0.0512	0.3348 ± 0.0578	**0.3819 ± 0.0291**
	NMI	0.1449 ± 0.0204	0.1351 ± 0.0242	0.1577 ± 0.0234	0.1050 ± 0.0377	0.1443 ± 0.0647	**0.1718 ± 0.0577**
	ARI	0.0691 ± 0.0134	0.0587 ± 0.0196	0.0567 ± 0.0129	0.0192 ± 0.0206	0.0586 ± 0.0595	**0.1000 ± 0.0580**
EthanolConcentration	ACC	0.2969 ± 0.0096	0.2970 ± 0.0126	0.2901 ± 0.0027	0.2953 ± 0.0189	0.2947 ± 0.0089	**0.2969 ± 0.0153**
	F1	0.2650 ± 0.0108	0.2696 ± 0.0121	0.2679 ± 0.0024	0.2705 ± 0.0203	0.2709 ± 0.0157	**0.2732 ± 0.0102**
	NMI	0.0110 ± 0.0032	0.0118 ± 0.0047	0.0138 ± 0.0044	0.0123 ± 0.0063	0.0176 ± 0.0044	**0.0190 ± 0.0157**
	ARI	0.0050 ± 0.0031	0.0050 ± 0.0039	0.0039 ± 0.0022	0.0055 ± 0.0060	0.0073 ± 0.0036	**0.0097 ± 0.0119**
FaceDetection	ACC	0.5054 ± 0.0042	0.5147 ± 0.0061	0.5095 ± 0.0066	0.5097 ± 0.0018	0.5135 ± 0.0060	**0.5198 ± 0.0044**
	F1	0.5053 ± 0.0041	0.5147 ± 0.0061	0.5083 ± 0.0065	0.5087 ± 0.0023	0.5135 ± 0.0061	**0.5192 ± 0.0046**
	NMI	0.0001 ± 0.0001	0.0007 ± 0.0005	0.0004 ± 0.0004	0.0003 ± 0.0001	0.0006 ± 0.0004	**0.0012 ± 0.0005**
	ARI	0.0001 ± 0.0002	0.0009 ± 0.0006	0.0004 ± 0.0006	0.0003 ± 0.0001	0.0008 ± 0.0006	**0.0015 ± 0.0006**
HandMovementDirection	ACC	0.2957 ± 0.0128	0.3171 ± 0.0272	0.3008 ± 0.0064	0.3167 ± 0.0144	0.3103 ± 0.0113	**0.3179 ± 0.0100**
	F1	0.2940 ± 0.0136	0.3146 ± 0.0306	0.2977 ± 0.0085	0.3127 ± 0.0129	0.3034 ± 0.0181	**0.3145 ± 0.0104**
	NMI	0.0138 ± 0.0031	0.0164 ± 0.0067	0.0129 ± 0.0067	0.0176 ± 0.0045	0.0171 ± 0.0047	**0.0208 ± 0.0050**
	ARI	-0.0015 ± 0.0028	0.0028 ± 0.0073	-0.0022 ± 0.0046	0.0031 ± 0.0082	0.0016 ± 0.0033	**0.0052 ± 0.0047**
Handwriting	ACC	0.2066 ± 0.0105	0.1510 ± 0.0073	0.2206 ± 0.0106	0.2432 ± 0.0084	0.2378 ± 0.0137	**0.2530 ± 0.0126**
	F1	0.2037 ± 0.0112	0.1479 ± 0.0073	0.2175 ± 0.0096	0.2549 ± 0.0162	0.2382 ± 0.0159	**0.2591 ± 0.0132**
	NMI	0.3000 ± 0.0205	0.1911 ± 0.0101	0.3242 ± 0.0181	0.3613 ± 0.0182	0.3439 ± 0.0180	**0.3652 ± 0.0139**
	ARI	0.0747 ± 0.0087	0.0273 ± 0.0039	0.0888 ± 0.0093	0.1080 ± 0.0125	0.1000 ± 0.0100	**0.1134 ± 0.0113**
Heartbeat	ACC	0.6637 ± 0.0059	0.6775 ± 0.0260	0.6440 ± 0.0372	0.6592 ± 0.0462	0.6805 ± 0.0075	**0.6817 ± 0.0546**
	F1	0.5227 ± 0.0909	0.5782 ± 0.0080	0.5538 ± 0.0574	0.5863 ± 0.0143	0.5871 ± 0.0225	**0.5944 ± 0.0367**
	NMI	0.0477 ± 0.0024	0.0418 ± 0.0043	0.0234 ± 0.0176	**0.0480 ± 0.0077**	0.0292 ± 0.0136	0.0382 ± 0.0220
	ARI	0.0360 ± 0.0631	0.0934 ± 0.0182	0.0436 ± 0.0287	0.0808 ± 0.0135	0.0872 ± 0.0182	**0.0981 ± 0.0485**
JapaneseVowels	ACC	0.7947 ± 0.0207	0.8238 ± 0.0666	0.6959 ± 0.0468	0.8419 ± 0.0418	0.8147 ± 0.0916	**0.8525 ± 0.0431**
	F1	0.8445 ± 0.0394	0.8241 ± 0.0589	0.7182 ± 0.0556	0.8158 ± 0.0271	0.8338 ± 0.0651	**0.8494 ± 0.0383**
	NMI	0.7897 ± 0.0276	0.7780 ± 0.0202	0.6570 ± 0.0370	0.7800 ± 0.0301	0.7723 ± 0.0517	**0.8113 ± 0.0180**
	ARI	0.6928 ± 0.0463	0.7244 ± 0.0650	0.4966 ± 0.0397	0.7141 ± 0.0602	0.6920 ± 0.0945	**0.7578 ± 0.0423**
LSST	ACC	0.1918 ± 0.0080	0.2014 ± 0.0194	0.2005 ± 0.0169	0.1930 ± 0.0133	0.1916 ± 0.0207	**0.2019 ± 0.0029**
	F1	0.1696 ± 0.0117	0.1556 ± 0.0133	0.1698 ± 0.0144	0.1680 ± 0.0069	0.1586 ± 0.0171	**0.1840 ± 0.0147**
	NMI	0.1452 ± 0.0088	**0.1527 ± 0.0111**	0.1248 ± 0.0129	0.1294 ± 0.0153	0.1107 ± 0.0213	0.1396 ± 0.0043
	ARI	0.0517 ± 0.0067	0.0472 ± 0.0061	0.0559 ± 0.0045	**0.0591 ± 0.0167**	0.0435 ± 0.0153	0.0549 ± 0.0028